

COLLEGE OF BUSINESS EDUCATION (CBE)

PROGRAMME: BIT 2

COURSE: INTERNET AND WEB DEVELOPMENT

ASSIGNMENT TYPE: INDIVIDUAL ASSIGNMENT

ISSUE DATE: 16/06/2026

PRESENTATION DATE: 16/07/2026

In the current digital era, organizations increasingly depend on web-based systems to automate operations, improve service delivery, and manage information efficiently. As a BIT student, you are required to demonstrate your ability to apply Object-Oriented Programming concepts and database management skills in developing a complete dynamic web application. Each student shall identify a real-world problem from any sector and design a web-based solution using PHP with Object-Oriented Programming principles and MySQL without using frameworks such as Laravel, CodeIgniter, Symfony, Yii, or CakePHP. The application should demonstrate the implementation of classes and objects, constructors, encapsulation, inheritance, polymorphism, abstraction, user authentication, session management, database connectivity using PDO or MySQLi, CRUD operations, form validation, search functionality, reporting features, and appropriate security practices. Students shall prepare system documentation including the problem background, objectives, requirements analysis, database design, screenshots of major interfaces, testing evidence, challenges encountered, and recommendations. All developed applications shall be deployed and hosted on each student's individual AWS account. During the presentation scheduled to commence on 16 July 2026, each student shall provide the live URL link to the hosted application and demonstrate its functionality while explaining the implementation of OOP concepts and responding to technical questions regarding the project. The use of copied source code, purchased projects, AI generated projects submitted without understanding and modification, or applications developed using prohibited frameworks shall result in penalties according to College regulations. Students are encouraged to demonstrate originality, creativity, professionalism, and adherence to coding standards throughout the development process. Additional technical requirements include designing the database in Third Normal Form (3NF) to minimize redundancy and maintain data integrity. Furthermore, all row data stored in database tables shall be encrypted before being persisted to the database, and appropriate mechanisms for secure decryption shall be implemented when retrieving authorized information. Students must clearly document the encryption approach, key management strategy, and demonstrate the effectiveness of the implemented security controls during the presentation.